Computational Project No. 1

1. Create a code for solving the following steady-state one-group transport problems in one-dimensional slab geometry:

   - the fixed-source problem

$$\mu\frac{\partial\psi(x,\mu)}{\partial x}+\Sigma_t(x)\psi(x,\mu)=\frac{1}{2}\Big(\Sigma_s(x)+\nu_f(x)\Sigma_f(x)\Big)\int_{-1}^{1}\psi(x,\mu')d\mu'+\frac{1}{2}q(x)\,, \tag{1a}$$

$$x\in[0,X]\,,\quad \mu[-1,1]\,,$$

$$\psi(0,\mu)=\begin{cases}\psi_{in}^{+}(\mu) & \text{for the boundary with incoming particles}\\ \psi(0,-\mu) & \text{for the reflective boundary}\end{cases}\quad \text{for } \mu>0\,, \tag{1b}$$

$$\psi(X,\mu)=\psi_{in}^{-}(\mu)\quad \text{for } \mu<0\,, \tag{1c}$$

   - the k-eigenvalue problem

$$\mu\frac{\partial\psi(x,\mu)}{\partial x}+\Sigma_t(x)\psi(x,\mu)=\frac{1}{2}\Big(\Sigma_s(x)+\frac{1}{k}\nu_f(x)\Sigma_f(x)\Big)\int_{-1}^{1}\psi(x,\mu')d\mu'\,, \tag{2a}$$

$$x\in[0,X]\,,\quad \mu[-1,1]\,,$$

$$\psi(0,\mu)=0\,,\quad \text{for } \mu>0\,, \tag{2b}$$

$$\psi(X,\mu)=0\quad \text{for } \mu<0\,. \tag{2c}$$

   Use the following numerical methods:

   - method of discrete ordinates (MDO) for angular discretization,
   - linear discontinuous scheme (LD) for spatial discretization,
   - the source iteration (SI) method for solving the discretized transport equation.

   For fixed-source problems, apply the convergence criterion given by

$$||\phi^{(\ell)}-\phi^{(\ell-1)}||_\infty<\varepsilon_\phi\left(\frac{1}{\rho_\phi^{(\ell)}}-1\right)\,, \tag{3}$$

   where ℓ is the index of source iterations, ε_ϕ is a parameter of the convergence criterion,

$$\rho_\phi^{(\ell)}=\frac{||\phi^{(\ell)}-\phi^{(\ell-1)}||_\infty}{||\phi^{(\ell-1)}-\phi^{(\ell-2)}||_\infty} \tag{4}$$

   is an estimation of the spectral radius of iterations.

For eigenvalue problems, the convergence criteria are defined by

$$||\phi^{(\ell)} - \phi^{(\ell-1)}||_\infty < \varepsilon_\phi \left(\frac{1}{\rho_\phi^{(\ell)}} - 1 \right) , \tag{5a}$$

$$\left| k^{(\ell)} - k^{(\ell-1)} \right| < \varepsilon_k \left(\frac{1}{\rho_k^{(\ell)}} - 1 \right) , \tag{5b}$$

where ε_k is a parameter,

$$\rho_k^{(\ell)} = \frac{|k^{(\ell)} - k^{(\ell-1)}|}{|k^{(\ell-1)} - k^{(\ell-2)}|} . \tag{6}$$

The resulting eigenfunction should be normalized such that

$$\int_0^a \phi(x) dx = 1 , \tag{7}$$

where a is the width of the slab.

Compute

- cell-average angular flux $\bar{\psi}_{m,i}$,
- cell-edge angular flux $\psi_{m,i+1/2}$,
- first Legendre spatial moment of the angular flux $\hat{\psi}_{m,i}$,
- cell-average scalar flux $\bar{\phi}_i$,
- cell-edge scalar flux $\phi_{i+1/2}$,
- first Legendre spatial moment of the scalar flux $\hat{\phi}_i$,
- cell-average current $\bar{J}_i$,
- cell-edge scalar flux $J_{i+1/2}$,
- first Legendre spatial moment of the current $\hat{J}_i$,
- scheme residuals on the ℓth iteration in mesh cells $(i = 1, \ldots, n_x)$
  - for the particle balance equation

$$\delta \bar{r}_i^{(\ell)} = J_{i+1/2}^{(\ell)} - J_{i-1/2}^{(\ell)} + \Sigma_{t,i} \bar{\phi}_i^{(\ell)} \Delta x_i - (\Sigma_{s,i} + \nu_{f,i} \Sigma_{f,i}) \bar{\phi}_i^{(\ell-1)} \Delta x_i - \bar{q}_i \Delta x_i , \tag{8a}$$

  - for the first spatial moment of the particle balance equation

$$\delta \hat{r}_i^{(\ell)} = 3(J_{i+1/2}^{(\ell)} + J_{i-1/2}^{(\ell)} - 2\bar{J}_i^{(\ell)}) + \Sigma_{t,i} \hat{\phi}_i^{(\ell)} \Delta x_i - (\Sigma_{s,i} + \nu_{f,i} \Sigma_{f,i}) \hat{\phi}_i^{(\ell-1)} \Delta x_i - \hat{q}_i \Delta x_i , \tag{8b}$$

- iterative residuals on the ℓth iteration in mesh cells $(i = 1, \ldots, n_x)$
  - for the particle balance equation

$$\delta \bar{R}_i^{(\ell)} = J_{i+1/2}^{(\ell)} - J_{i-1/2}^{(\ell)} + (\Sigma_{t,i} - \Sigma_{s,i} - \nu_{f,i} \Sigma_{f,i}) \bar{\phi}_i^{(\ell)} \Delta x_i - \bar{q}_i \Delta x_i , \tag{9a}$$

  - for the first spatial moment of the particle balance equation

$$\delta \hat{R}_i = 3(J_{i+1/2}^{(\ell)} + J_{i-1/2}^{(\ell)} - 2\bar{J}_i^{(\ell)}) + (\Sigma_{t,i} - \Sigma_{s,i} - \nu_{f,i} \Sigma_{f,i}) \hat{\phi}_i^{(\ell)} \Delta x_i - \hat{q}_i \Delta x_i . \tag{9b}$$

2. Solve the transport problems described below.

- **Test A**

  Consider a 1D slab ($x \in [0, 8]$ *cm*) with five regions made of different materials. The parameters of the problem are listed in Table 1. The left boundary is reflective. The right boundary is vacuum. Use

  – a uniform spatial mesh with $n_x = 40$ cells, $\Delta x = \frac{8}{n_x}$,
  – S_8 Gauss-Legendre quadrature set ($n_\mu = 8$),
  – the convergence criterion (3) with $\varepsilon_\phi = 10^{-6}$.

Table 1: Parameters of Test A

Subdomain	1	2	3	4	5
Right boundary of the subdomain [cm]	2	3	5	6	8
Σ_t [cm^{-1}]	50	5	0	1.1	1.1
Σ_s [cm^{-1}]	0	0	0	0.7	0.5
Σ_f [cm^{-1}]	0	0	0	0.1	0.16
ν_f [neutrons per fission]	0	0	0	2	2.5
q $[neutrons/(cm^3\ sec)]$	50	0	0	0	1

- **Test B**

  – There are two spatial regions in the problem:
    * $\mathcal{A}_1$: $x \in [0, 1]$ *cm*, $\Sigma_t = 2$ cm^{-1}, $\Sigma_s = 0$, $\Sigma_f = 0$, $q = 0$,
    * $\mathcal{A}_2$: $x \in [1, 11]$ *cm*, $\Sigma_t = 100$ cm^{-1}, $\Sigma_s = 100$ cm^{-1}, $\Sigma_f = 0$, $q = 0$.
  – The left boundary conditions is given by $\psi(0, \mu) = 1$ for $\mu > 0$.
  – The right boundary condition is vacuum.
  – The spatial mesh in the subdomains is defined with the following intervals:
    * $\mathcal{A}_1$: $\Delta x = 0.1$,
    * $\mathcal{A}_2$: $\Delta x = 1$.
  – Use the S_8 Gauss-Legendre quadrature set ($n_\mu = 8$),
  – the convergence criterion (3) with $\varepsilon_\phi = 10^{-6}$.

- **Test C**

  Given a slab $0 \leq x \leq a$ cm having $\Sigma_t = 1$ cm^{-1}, $\Sigma_s = 0.45$ cm^{-1}, $\Sigma_f = 0.3\,cm^{-1}$, $\nu_f = 2\,\frac{neutrons}{fission}$,. The boundary conditions are vacuum. Use

  – a uniform spatial mesh with $n_x = 50$ cells, $\Delta x = \frac{a}{N}$,
  – S_8 Gauss-Legendre quadrature set.

  Calculate the critical width of the slab, a_{crit}.

3. In your report you need to present
   - the algorithm flowchart,
   - numerical results for each test in the following form:
     - for Test A & B
       * number of source iterations N_{iter},
       * graph of $||\phi^{(\ell)} - \phi^{(\ell-1)}||_\infty$ versus ℓ,
       * graph of $\rho_\phi^{(\ell)}$ versus ℓ,
     - for Test C
       * the obtained value of critical width ,
       * graph of iterative estimations of the critical width $a_{crit}^{(s)}$ versus s, where s is the index of iterations for the critical width calculations,
     - the following plots of obtained numerical solutions in each test (i.e. Test A, B & C):
       * the cell-average values of the scalar flux, ϕ_i, versus position x_i ,
       * the first spatial moment of the scalar flux, $\hat{\phi}_i$, versus position x_i
       * the cell-edge values of the scalar flux, $\phi_{i+1/2}$, versus position $x_{i+1/2}$,
       * the cell-edge values of the current, $J_{i+1/2}$, versus position $x_{i+1/2}$,
       * the cell-average values of the angular flux, $\bar{\psi}_{m,i}$, versus (x_i, μ_m),
       * the residuals $\delta\bar{r}_i$, $\delta\hat{r}_i$, $\delta\bar{R}_i$, $\delta\hat{R}_i$ versus position x_i.
   - files with your code.